

IRA WANG

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Education

Bachelor of Engineering (Honours) — Electrical & Electronic Engineering

University of Auckland | Activities: IEEE (Executive Team), Women in Engineering, Investment Club, UoA Social Volleyball Club

Mar 2024 - Nov 2027 (expected)

Exchange Programme, Electrical Engineering

National Taiwan University

Jan – Jun 2026

Relevant Coursework

Analogue & Digital Design	PCB layout in Altium Designer; LTspice & Proteus simulation; embedded C for ATmega328P in Microchip Studio; soldering.
Electronics	Op-amp circuits, BJT/MOSFET transistors, frequency response, Bode plots.
Electromagnetics	Electrostatics, magnetostatics, transmission lines; finite-element analysis with FEMM.
Innovation & Business	Innovation process, business case development, project management.
Professional Skills	Engineering communication, ethics, sustainability, report writing.
Engineering Computation	C, MATLAB, OOP principles, data structures and algorithms in Java.

Projects

Garbage Bag Recognition System

Part III Project

- Identified unsorted garbage disposal as a key driver of cleaning staff workload through stakeholder interviews and root-cause analysis.
- Built a custom image dataset and trained a YOLOv8 object-detection model to identify improperly disposed garbage bags in real time.
- Deployed the trained model on a Raspberry Pi, integrating a live camera feed with a web dashboard for evidence review and reporting.
- Used AI tools throughout development to accelerate model iteration, debug training pipelines, and improve detection accuracy.

Skills gained: Gemini, Copilot, Cursor, YOLOv8, Python, Raspberry Pi.

Energy Monitor

Part II Project

- Designed and routed a custom PCB in Altium Designer to measure voltage, RMS current, and average power; output to a 7-segment display module.
- Wrote embedded C firmware for an ATmega328P in Microchip Studio; validated signal behaviour in LTspice and Proteus.
- Soldered and assembled the finished board, then used an oscilloscope to debug and verify waveforms.

Skills gained: Altium Designer, LTspice, Proteus, Microchip Studio (C), soldering iron, oscilloscope.

SMART Car

IEEE Outreach Event

- Organised and led a hands-on workshop introducing first-year students to electrical engineering by building an autonomous SMART car.
- Programmed obstacle-avoidance logic in Arduino C++ using ultrasonic sensors, and guided participants through wiring, assembly, and testing.

Skills gained: Arduino C++, sensor integration, workshop facilitation, technical mentoring.

Work Experience

Wafer Test Intern

Dec 2025 – Feb 2026

NXP Semiconductors — Kaohsiung, Taiwan

- Supported semiconductor wafer test operations, using RPA, KVM, and VNC to streamline data review and reduce manual testing time.
- Integrated AI tools into lab workflows to improve testing efficiency and accuracy across high-volume wafer review processes.
- Improved accuracy by 80% after the first milestone.

- Authored an SOP handbook standardising RPA testing procedures, improving process consistency and reducing errors across the team.
- Presented test progress, process improvements, and project timelines (Gantt charts) to senior engineers.

Skills gained: *process automation, technical documentation, stakeholder communication.*

Waitress

Aug 2024 – Jun 2025

QT Hotel — Esther Restaurant, Auckland

- Delivered attentive, precise service and resolved customer inquiries in a fast-paced dining environment.

Skills gained: *customer service, multitasking, communication under pressure.*

Tutor

Feb 2020 – Current

Private Tutoring, Auckland

- Tutored children aged 6–11 in Math, English, and Chinese.

Skills gained: *patience, adaptability, clear communication.*

Technical Skills

Languages	C, Java, MATLAB
EDA / CAD	Altium Designer (PCB layout & routing), LTspice, PLECS, Proteus, Intel Quartus, CAD (Inventor)
Embedded	ATmega328P, Microchip Studio, SMT soldering, PCB design, oscilloscope signal analysis
Software	Git, Excel, MATLAB
Soft Skills	Team leadership, technical documentation, project coordination

Volunteer Experience

- Westlake Girls High School UNICEF Committee — Organised fundraising events (quiz night, bake sale) advocating for children's welfare; raised \$3,000 (2023).
- Chinese Night, Child Cancer Foundation — Organised a cultural performance, securing venues and sponsorship and coordinating production; raised \$4,000.

Hobbies

- Home Renovation
- Learning French
- Swimming
- Travelling

Last updated August 2026